# Market Dashboard Expansion

## Claude Code Handover Document

Version 1.0 | March 2026 | Prepared for handover from claude.ai to Claude Code

| Property | Value |
|---|---|
| Repository | github.com/Kasim81/market_dash_auto |
| Google Sheet ID | 12nKIUGHz5euDbNQPDTVECsJBNwrceRF1ymsQrIe4_ac |
| Primary contact | kasimzafar@gmail.com |
| Existing secrets | FRED_API_KEY, GOOGLE_CREDENTIALS |
| New secrets needed | BLS_API_KEY, FMP_API_KEY |
| Python version | 3.11 (set in update_data.yml) |
| Run schedule | Daily 06:00 UTC via GitHub Actions |

# 1. What Currently Exists (Do Not Break)

The following files, tabs, and outputs are live and in production. Claude Code must never modify these without explicit instruction.

## 1.1 Repository Files

| File | Purpose | Status |
|---|---|---|
| fetch_data.py | Main daily collection: 66 instruments, FX, FRED yields/sentiment, Google Sheets push | PRODUCTION — do not modify logic |
| fetch_macro_us_fred.py | 25 US FRED macro indicators → macro_us tab (snapshot, now being replaced by history) | PRODUCTION |
| fetch_hist.py | NEW — historical weekly time series for market_data_hist and macro_us_hist tabs | JUST DEPLOYED — verify first run |
| requirements.txt | yfinance, pandas, numpy, requests, google-auth, google-api-python-client | Add new packages here |
| .github/workflows/update_data.yml | GitHub Actions: daily 06:00 UTC, injects secrets, commits CSVs | Do not change schedule without instruction |

## 1.2 Google Sheet Tabs

| Tab Name | GID | Written By | Contents |
|---|---|---|---|
| market_data | 68683176 | fetch_data.py | 75 rows: 66 instruments + 5 ratios + headers. Daily snapshot of prices and returns (1W/1M/3M/6M/YTD/1Y, local + USD). NEVER TOUCH. |
| sentiment_data | 14812133 | fetch_data.py | 5 FRED sentiment series: UMich, Conference Board, ISM, etc. NEVER TOUCH. |
| macro_us | (new) | fetch_macro_us_fred.py → being replaced | Was snapshot of 25 US macro indicators. Being replaced by macro_us_hist (full history). |
| market_data_hist | (new) | fetch_hist.py | Weekly Friday prices from 2000-01-07. 1,366 rows × 144 cols. All Local prices then all USD prices. |
| macro_us_hist | (new) | fetch_hist.py | Weekly Friday series from 1947-01-01. 4,132 rows × 26 cols. 25 FRED series, forward-filled to weekly. |

| | | | |
|---|---|---|---|
| Market Data | duplicate | Manual / formula | REDUNDANT — duplicate of market_data. Can be deleted after confirming it contains no unique formulas. |

## 1.3 CSV Export URLs (used by trigger.py)

```
market_data:
https://docs.google.com/spreadsheets/d/12nKIUGHz5euDbNQPDTVECsJBNwrceRF1ymsQrIe4_ac/export?format=csv&gid=68683176
```

```
sentiment_data:
https://docs.google.com/spreadsheets/d/12nKIUGHz5euDbNQPDTVECsJBNwrceRF1ymsQrIe4_ac/export?format=csv&gid=14812133
```

Note: trigger.py lives at C:\Users\kasim\ClaudeWorkflow\ and runs at 06:15 London time. History tabs are for manual analysis only — trigger.py does not read them.

## 2. Schema Decisions (Locked)

These decisions were made in the planning session and must not be revisited without explicit instruction.

| Decision | Choice Made | Rationale |
|---|---|---|
| Data orientation | Long format: dates as rows, indicators as columns | 10M row limit vs 18,278 col limit — long format scales better |
| market_data_hist date range | 2000-01-07 to present (Fridays) | Most instruments have reliable data from 2000 |
| macro_us_hist date range | 1947-01-01 to present (Fridays) | Earliest FRED data; series fill in as they become available |
| Frequency | Weekly, Friday close (W-FRI resample) | Consistent spine; daily would be 5× the rows with marginal benefit |
| Monthly macro in weekly sheet | Forward-fill into intervening weekly slots | Clean spine; blanks rejected because they make charting harder |
| Local vs USD grouping | All Local rows first, then all USD rows | Grouped separately, not interleaved |
| macro_us snapshot tab | Replace with history — drop snapshot | Snapshot was intermediate artefact; history supersedes it |
| trigger.py and history tabs | History tabs are for manual analysis only | trigger.py reads market_data and sentiment_data only |

# 3. Source Architecture

| Layer | Coverage | API Key? | Status |
|---|---|---|---|
| yfinance (existing) | All market prices, FX, ETFs, yields (66 instruments) | None | Production |
| FRED API (existing) | US macro: breakevens, yield curve, M2, SLOOS, payrolls, CPI, PCE (25 series) | FRED_API_KEY | Production |
| OpenBB → OECD (new) | Multi-country: CLI, CPI, unemployment, GDP, interest rates | None needed | Phase C |
| OpenBB → IMF (new) | Multi-country: GDP, CPI, current account, EM coverage | None needed | Phase C |
| OpenBB → ECB (new) | Eurozone: ECB policy rate, M3, bank lending | None needed | Phase C |
| OpenBB → BLS (new) | US: nonfarm payrolls detail | BLS_API_KEY (free) | Phase B |
| FMP (new, narrow) | ISM Manufacturing, ISM Services, Philly Fed, Empire State only | FMP_API_KEY (free) | Phase D |

## New pip installs required

```
openbb openbb-oecd openbb-imf openbb-ecb openbb-federal-reserve openbb-bls openbb-fmp
openbb-fred
```

Add all to requirements.txt. Test locally before committing.

# 4. Remaining Implementation Phases

Phases are ordered by priority. Each phase is independent — a failure in one should not block others.

## Phase B — OpenBB Survey Endpoints

Module: fetch_macro_surveys.py

Output tab: macro_surveys (new)

Indicators to fetch:

- SLOOS detail — net tightening by loan category (via OpenBB Federal Reserve)
- Regional Fed surveys — Philly Fed Mfg, Empire State Mfg (via FMP as fallback)
- UMich consumer sentiment detail — current conditions vs expectations (FRED: UMCSENT already in macro_us; need sub-indices)

Rate limiting: match FRED pattern (0.6s delay, exponential backoff on 429).

## Phase C — International Macro (OpenBB OECD / IMF / ECB)

Module: fetch_macro_international.py

Output tab: macro_international (new)

Countries and regions:

| Region | Countries | ISO Codes |
|---|---|---|
| North America | United States, Canada | US, CA |
| UK | United Kingdom | GB |
| Europe | Germany, France, Eurozone aggregate | DE, FR, EZ |
| Japan | Japan | JP |
| Asia ex-Japan | China, India, South Korea, Australia | CN, IN, KR, AU |

Indicators per country (where available):

- OECD CLI (Composite Leading Indicator) — via OpenBB OECD
- CPI YoY — via OpenBB OECD or IMF
- Unemployment rate — via OpenBB OECD
- GDP YoY — via OpenBB IMF
- Policy interest rate — via OpenBB ECB (Eurozone) or OECD (others)

Schema: long format. Columns: Date | {indicator}_{ISO} (e.g. CLI_US, CLI_DE, CPI_JP). Forward-fill monthly to weekly.

## Phase D — FMP for PMI / Survey Data

Module: fetch_pmi.py (or integrate into fetch_macro_surveys.py)

Output: append columns to macro_us_hist or new macro_pmi tab

FMP free tier covers:

- ISM Manufacturing PMI (US) — NAPMPI series
- ISM Services PMI (US) — NMI series
- Philadelphia Fed Manufacturing Survey
- Empire State Manufacturing Survey (NY Fed)

Free API key: financialmodelingprep.com. Store as FMP_API_KEY secret.

## Phase E — New ETFs (Instrument Basket Expansion)

Additions to fetch_data.py instrument lists. No new modules needed.

Categories to expand:

- Europe sector ETFs (.DE listed, EUR currency)
- EM regional ETFs (Latin America, Africa, EMEA)
- UK style ETFs (quality, momentum factors)
- Asia/Japan additional coverage

For each new LSE (.L) ETF: verify pence correction logic applies. For each new non-USD instrument: verify currency is in FX_TICKERS dict.

Confirm full list with Kasim before implementing — instrument selection is owner decision.

## Phase F — Calculated Fields

Append synthetic columns to market_data_hist and macro_us_hist after all instrument fetches.

| Field | Formula | Signal |
|---|---|---|
| HY/IG ratio | BAMLH0A0HYM2 / BAMLC0A0CM (from FRED) | Credit risk appetite; compression = risk-on |
| EMFX basket | Equal-weight CNY, INR, KRW, TWD vs USD | EM risk appetite proxy |
| EEM/IWDA ratio | EEM close / IWDA.L close (FX-adjusted) | EM vs DM relative momentum |
| MOVE proxy | 30-day realised vol on ^TNX (10Y yield) | Bond volatility proxy; substitute for MOVE index |
| Global PMI proxy | Equal-weight ISM + Eurozone PMI + Japan PMI | Global growth signal |
| Global yield curve | Average of US/DE/UK/JP 10Y-2Y spreads | Synchronised monetary tightening signal |

## Phase G — Google Sheets Export Updates

Ensure all new tabs are pushed by the relevant fetch modules. Verify:

- Tab names match exactly (lowercase underscores)
- Protected tabs guard is in place: never write to market_data or sentiment_data
- Batch write logic (10,000 rows per call) is used for large tabs
- GIDs are recorded in this document once tabs are created

## 5. Indicator Reference — US Macro (25 FRED Series)

These are already implemented in fetch_macro_us_fred.py and fetch_hist.py. Listed here for reference when building Phase D / F cross-references.

| FRED Series | Name | Category | Frequency |
|---|---|---|---|
| T10Y2Y | US Yield Curve 10Y-2Y Spread | Growth / Monetary | Daily |
| T10Y3M | US Yield Curve 10Y-3M Spread | Growth / Monetary | Daily |
| M2SL | US M2 Money Supply | Monetary | Monthly |
| USSLIND | US Conference Board LEI | Growth | Monthly |
| PERMIT | US Building Permits (SAAR) | Growth | Monthly |
| IC4WSA | Initial Jobless Claims 4-Week Avg | Growth | Weekly |
| PAYEMS | US Nonfarm Payrolls | Growth | Monthly |
| UNRATE | US Unemployment Rate | Growth | Monthly |
| INDPRO | US Industrial Production | Growth | Monthly |
| RSXFS | US Retail Sales ex-Autos | Growth | Monthly |
| DRTSCILM | SLOOS Net Tightening (C&I Large) | Financial Conditions | Quarterly |
| NFCI | Chicago Fed NFCI | Financial Conditions | Weekly |
| CPIAUCSL | US CPI Headline | Inflation | Monthly |
| CPILFESL | US Core CPI | Inflation | Monthly |
| PCEPILFE | US Core PCE Deflator | Inflation | Monthly |
| PPIACO | US PPI All Commodities | Inflation | Monthly |
| T5YIE | US TIPS 5Y Breakeven | Inflation | Daily |
| T10YIE | US TIPS 10Y Breakeven | Inflation | Daily |
| T5YIFR | US 5Y5Y Forward Inflation | Inflation | Daily |
| MICH | UMich Consumer Inflation Expectations | Inflation | Monthly |
| FEDFUNDS | US Federal Funds Rate | Monetary | Monthly |
| DFII10 | US 10Y Real Rate (TIPS) | Monetary | Daily |
| DFII5 | US 5Y Real Rate (TIPS) | Monetary | Daily |
| BAMLH0A0HYM2 | US HY Credit Spread OAS | Financial Conditions | Daily |
| BAMLC0A0CM | US IG Credit Spread OAS | Financial Conditions | Daily |

## 6. Indicators Excluded (With Rationale)

Do not attempt to implement these without explicit instruction. The reasons are noted to avoid re-researching dead ends.

| Indicator | Why Excluded | Proxy Used Instead |
|---|---|---|
| Atlanta Fed GDPNow | Web scrape only — no clean API endpoint | None (GDP advance from BEA via FRED: GDPC1) |
| Fed Funds Futures implied path | CME data — paid subscription required | None (omitted from Phase 1 scope) |
| Taylor Rule Gap | Requires custom calculation from multiple inputs | Deferred to Phase 2 content workflow |
| Global Monetary Policy Tracker | BIS/JP Morgan data — no free API | None (omitted) |
| NAHB Housing Market Index | Low priority vs Building Permits (already covered) | PERMIT (FRED) covers housing leading signal |
| Goldman Sachs FCI | Proprietary — not publicly available | Chicago Fed NFCI (FRED: NFCI) |
| Bloomberg FCI | Bloomberg terminal required | Chicago Fed NFCI |
| MOVE Index | Bloomberg terminal required | 30-day realised vol on ^TNX (Phase F) |
| JP Morgan Global PMI | S&P Global licence required | Equal-weight ISM + EZ PMI + Japan PMI (Phase F) |
| EM Currencies Index (EMFX) | JP Morgan index — proprietary | Basket of CNY/INR/KRW/TWD vs USD (Phase F) |
| China 10Y yield (FRED) | Series CHNGDPNQD — data quality issues | CNYB.L ETF used as proxy |
| UK 2Y yield (FRED) | Series IUDSGT02 — retired/invalid | ^IRX used as proxy if needed |

# 7. Coding Conventions to Preserve

All new modules must follow these patterns — they exist in production code and deviations will cause inconsistency.

## 7.1 Rate Limiting

- yfinance: 0.3s delay between per-ticker calls (YFINANCE_DELAY = 0.3)
- FRED: 0.6s delay between calls (FRED_DELAY = 0.6)
- FRED exponential backoff on 429/5xx: 2s → 4s → 8s → 16s → 32s (max 5 retries)
- yfinance: prefer bulk download (yf.download([list])) over per-ticker loops

## 7.2 Error Handling

- Per-series try/except — one failure must never kill the run
- Broad try/except wrapper in fetch_data.py integration block — Phase failures cannot affect existing pipeline
- If FRED_API_KEY absent: skip FRED fetches silently, log warning, continue
- If GOOGLE_CREDENTIALS absent: skip Sheets push, log warning, continue

## 7.3 LSE Pence Correction

```
if ticker.endswith('.L') and median_price > 50: price /= 100
```

Applied to all .L tickers. Median check prevents double-correction on re-runs.

## 7.4 USD Return Calculation

```
usd_return = (1 + local_return) * (1 + fx_return) - 1  # compounded, not additive
```

FX direction convention:

- GBPUSD=X, EURUSD=X: multiply (USD is quote currency — rising pair means GBP/EUR strengthened)
- USDJPY=X, USDCNY=X, USDINR=X, USDKRW=X, USDTWD=X: divide (USD is base currency)

## 7.5 Google Sheets Push

- Protected tab guard: hard-refuse writes to market_data and sentiment_data
- Batch write: 10,000 rows per API call to stay within payload limits
- NaN → empty string before push (never push 'nan' strings)
- Tab creation: check existing tabs before creating; use batchUpdate addSheet request
- Clear before write: sheets.values().clear() on the full range before updating

### 7.6 CSV Commit Logic

fetch_data.py uses a diff-check before committing CSVs — only commits if content changed. Preserve this in new modules to avoid noisy daily commits when data hasn't changed (e.g. weekends).

# 8. First Steps for Claude Code

When starting a new Claude Code session on this project, follow this sequence:

## Step 1 — Verify the current state

```
cat data/market_data_hist.csv | head -3      # confirm hist fetch ran
cat data/macro_us_hist.csv | head -3         # confirm macro hist ran
python fetch_hist.py                          # test run if needed
```

## Step 2 — Check which GitHub secrets are present

Go to Settings → Secrets → Actions in the repo. Confirm: FRED_API_KEY, GOOGLE_CREDENTIALS. Note which are missing: BLS_API_KEY, FMP_API_KEY (needed for phases B and D).

## Step 3 — Install and test OpenBB

```
pip install openbb openbb-oecd openbb-imf openbb-ecb openbb-federal-reserve openbb-bls openbb-fred
python -c "from openbb import obb; data = obb.economy.indicators(countries=['US'], indicators=['CLI']); print(data)"
```

If OpenBB install fails or OECD endpoints return empty, fall back to direct OECD API (data.oecd.org/api/v2) — no key required.

## Step 4 — Decide on Phase B or Phase C first

Phase B (survey endpoints) is simpler and self-contained. Phase C (international macro) has more complexity due to multi-country alignment. Recommended order: B → C → D → E → F → G.

## Step 5 — Test any new module before integrating into fetch_data.py

```
python fetch_macro_surveys.py    # run standalone first
```

Only add the integration call to fetch_data.py after standalone test passes. This matches how fetch_hist.py was deployed.

# 9. Known Issues and Watch Points

| Issue | Status | Action |
|---|---|---|
| 'Market Data' tab (space) is duplicate of market_data (underscore) | Identified — not yet deleted | Delete 'Market Data' tab after confirming it has no unique formulas |
| macro_us snapshot tab | Being replaced by macro_us_hist | Delete macro_us once macro_us_hist is confirmed working |
| fetch_hist.py first run takes 8–12 mins | Expected — fetching decades of history | Subsequent daily runs will be fast (only new week appended) |
| China 10Y FRED series removed | CHNGDPNQD had data quality issues | CNYB.L ETF used as proxy in market basket |
| UK 2Y FRED series removed | IUDSGT02 series ID retired | ^IRX used as proxy if needed |
| NAPMPI vs NAPM | NAPM retired; NAPMPI is current ISM series | Already fixed in production — do not revert |
| GitHub Actions inactivity pause | Actions pauses after 60 days of no pushes | Push trivial commit or re-enable from Actions tab if pipeline stops |
| Google Sheets CDN caching | GitHub raw CSV URL caches aggressively | Always use Sheets export URL (gid= parameter) not GitHub raw URL |

# 10. Quick Reference

## Key URLs

```
Repo:          https://github.com/Kasim81/market_dash_auto
Sheet:
https://docs.google.com/spreadsheets/d/12nKIUGHz5euDbNQPDTVECsJBNwrceRF1ymsQrIe4_ac
market_data:   ...export?format=csv&gid=68683176
sentiment_data: ...export?format=csv&gid=14812133
FRED API:      https://api.stlouisfed.org/fred/series/observations
FRED register: https://fred.stlouisfed.org → My Account → API Keys
FMP register:  https://financialmodelingprep.com/developer
BLS register:  https://www.bls.gov/developers/api_signature_v2.htm
```

## Secret Names (GitHub repo → Settings → Secrets → Actions)

| Secret Name | Used For | Status |
|---|---|---|
| FRED_API_KEY | All FRED API calls | Exists |
| GOOGLE_CREDENTIALS | Google Sheets push (full service account JSON) | Exists |
| BLS_API_KEY | BLS nonfarm payrolls detail (Phase B) | MISSING — register at bls.gov |
| FMP_API_KEY | ISM/PMI data via Financial Modeling Prep (Phase D) | MISSING — register at financialmodelingprep.com |